

US FLIGHT PERFORMANCE ANALYSIS DASHBOARD

Project Report

This report presents a Flight Performance Analysis Dashboard developed using Power BI and PostgreSQL. The dashboard analyzes flight operations, airline performance, airport activity, delays, and cancellations across the United States. Interactive visualizations and KPIs help identify operational inefficiencies and performance trends. The insights generated support data-driven decisions to improve service reliability and overall flight operations.

1. Project Overview

- **Project Title:** Flight Performance Analysis Dashboard
- **Domain:** Aviation / Airline Operations
- **Tool Used:** Microsoft Power BI, PostgreSQL
- **Data Source:** US Flight Performance Dataset
- **Datasets Used:** Flights, Airlines, Airports
- **Dashboard Pages:**
 - Flight Performance Dashboard
 - Airline Performance Dashboard
 - Airport Analysis Dashboard

Key KPIs

- Total Flights
- Cancelled Flights
- OTP Rate (%)
- Average Arrival Delay
- Average Departure Delay
- Cancellation Rate (%)

2. Business Problem Statement

Airlines generate millions of flight records every year. Monitoring delays, cancellations, airline performance, and airport operations manually is difficult and time-consuming.

The objective of this project is to transform raw flight data into interactive dashboards that help identify operational inefficiencies, delay patterns, cancellation trends, and airport performance issues. These insights support data-driven decision making and operational improvement.

3. Business Objectives

- Analyze overall flight performance.
- Monitor airline-wise operational efficiency.
- Identify airports with high delays and cancellation rates.
- Measure On-Time Performance (OTP).
- Track flight volume and cancellation trends.
- Generate actionable insights to improve airline operations and passenger experience.

4. Data Collection and Preprocessing

The data used in this project consists of US flight operations data including airline, airport, delay, and cancellation information.

Data preparation activities included:

- Importing data into PostgreSQL.
- Data cleaning and validation.
- Handling missing values.
- Standardizing airport and airline codes.
- Creating calculated measures using DAX.
- Building relationships between Flights, Airlines, and Airports tables.

5. Tools and Technologies Used

- **Power BI Desktop** – Dashboard development and visualization.
- **PostgreSQL** – Data storage and SQL analysis.
- **Power Query** – Data transformation and cleaning.
- **DAX** – KPI and measure creation.
- **Microsoft Excel** – Data validation and review.

6. Dashboard Features

Flight Performance Dashboard

- Monthly Flight Trend
- Top Airlines by Flight Count
- Cancellation Reason Analysis
- Top Origin Airports
- KPI Tracking

Airline Performance Dashboard

- OTP Rate by Airline
- Average Arrival Delay
- Average Departure Delay
- Top Airlines by Flights
- Airline-wise Performance Comparison

Airport Analysis Dashboard

- Top Airports by Average Distance
- Top Origin Airports by Flights
- Top Destination Airports by Flights
- Airport Cancellation Rate Analysis
- Airport Performance Tracking

7. Dashboard Screenshots:



8. Key Business Insights

- More than **6 Million flights** were analyzed.
- Overall cancellation rate remained around **1.54%**.
- Weather was identified as the leading cancellation reason.
- Certain airlines achieved significantly higher OTP rates than others.
- Some airports experienced higher cancellation percentages, indicating operational challenges.
- Major hub airports handled the highest flight traffic.
- Delay patterns varied across airlines and airports, highlighting opportunities for operational improvement.

9. Challenges Faced

- Handling a large flight dataset.
- Cleaning inconsistent airport and airline records.
- Creating accurate OTP and Cancellation Rate measures.
- Designing interactive dashboards with meaningful business insights.
- Optimizing report performance for large volumes of data.

10. Conclusion

The Flight Performance Analysis Dashboard successfully transformed raw flight operation data into meaningful business insights. The analysis helped identify key factors affecting airline and airport performance, including delays, cancellations, and operational efficiency.

The dashboard highlights airports and airlines requiring performance improvement while providing visibility into flight trends and service reliability. These insights can support better scheduling, resource planning, and decision-making to improve overall passenger experience and operational performance.

11. Future Enhancements

- Integration of real-time flight data.
- Predictive delay forecasting using machine learning.
- Route-wise performance analysis.
- Weather impact analysis.
- Advanced airport congestion monitoring.
- Mobile-friendly dashboard deployment.